[G] Origin of New Species

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The Origin of new species

- ecological isolation (physical barriers)
- reproductive isolation
  - different mating schedule
  - different sexual behaviors
  - mechanical differences
- incompatible gametes
- hybrid inviability: fertilization possible but zygotes do not survive
- hybrid sterility: hybrid species can breed but offspring cannot

isolation creates new species

Gene Selection

- isolated populations will evolve differently (genetic drift)
- more separation ⇒ more incompatibility ⇒ less breeding ⇒ new species

Adaptive Radiation

- when subpopulations adapt to new ecological niches differently
- 3 circumstances:
  1. no similar/competing species
  2. new environment
     - mass extinction
     - migration
  3. species is <u>generalized</u>
     - already has multiple skills/features and can easily survive

③ species is generalized

- already has multiple skills/features and can easily survive in new environments
  - ↳ example: humans

Darwinian Gradualism

- the tree of evolution
- lots of little branches diverge, but main branch slowly, gradually changes to new species (microevolution)
- challenged by modern fossil record
  - species are stable for a long time, then suddenly change
  - not gradual change, but macroevolution/punctuated equilibrium
  - natural selection is like an 'editor' of species, while genetic drift/flow/mutation are the 'writers'

REVIEW

- Mutation is only new variation
- Gene flow/drift mix variation
- Natural Selection shapes variation
- Segregation, independent assortment allow for new combinations